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Preventing Fluid Leaks to the Ocean from Slip-Joint Packers with Digital Subsea Technology

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Abstract

On floating drill ships, semi-submersible rigs and platforms, slip joint packers are used to seal drilling fluid within the telescopic riser. These elastomer seals are subject to wear through the action of ocean swell and tide and can eventually leak. If the elastomer seal failure/leak is undetected, it can result in drilling/circulating fluid spillage to the ocean and risk damage to the environment. Given the potential for reputational damage and significant expense, this is a situation that must be avoided and de-risked during operations.

New packer management technology has been developed to immediately detect and eliminate fluid leaks to the environment. If the rig slip-joint packer fails, the packer management system immediately detects the pressure drop, and activates the back-up packer to avoid a spill to the ocean. As a rig/floating platform slip-joint packer wears, packer pressure must be steadily increased to maintain a positive seal. The system applies intelligent digital 'dead band' tracking to automatically adjust the alarm 'trip point'. Allowing near-instantaneous reaction time to packer leaks. Both the active packer and the back-ups are monitored, ensuring that the back-up is fully operational and immediately available. Multiple slip-joint packers can be simultaneously monitored. Ensuring that a further back-up is always ready and available, thus avoiding down time while a packer seal is replaced during a drilled interval.

The system is automated, intelligent and removes the human factor, reacting near-instantaneously to prevent spills and environmental incidents. Increasing the sustainability of sub-sea field development for operators. Slip-joint packer operation is monitored 24/7 and data can be transmitted and accessed live. In the event of an incident operators have immediate access to data. If problems are encountered, these may be detected and solved remotely, with the number of rig visits significantly reduced.

Introduction

As older shallow offshore and land fields have depleted, the oil and gas industry has rapidly moved towards the development of deeper offshore reservoirs. The development of sub-sea technology has enabled the significant growth of deepwater drilling activity in areas such as the Gulf of Mexico, West Africa, Brazil, Asia Pacific and the development of new ultra deepwater frontiers such as Guyana. While the importance of deepwater fields has grown, industry has faced increasing environmental scrutiny from both government regulators and the public. National offshore regulators such as the National Offshore Petroleum Safety and Environmental Management Authority (NOPSEMA) in Australia, the Offshore Petroleum Regulator for Environment and Decommissioning (OPRED) in the UK and the United States Environmental Protection Agency (EPA) demand that all offshore spills must be reported. Failure to report accidental spills can constitute an offence, with some regulators wielding the power to suspend offshore activity or in some circumstances suspend licenses. It is evident that the development of sub-sea technology must continue to meet this challenge to enable the continued exploitation of important new deepwater and ultra deepwater plays.

Semi-submersible rigs, drill ships and floating production platforms are widely applied to explore, develop and produce deep water reservoirs worldwide. Circulating fluid is sealed within the telescopic riser on floating rigs and platforms using a slip joint packer assembly. The telescopic riser allows the floating rig to move with the action of heave and tide. However, this movement also causes frictional wear to the slip joint packer sealing element. This wear will eventually cause the slip joint packer to leak circulating fluid into the ocean. Potentially resulting in an environmental incident for the operator and rig/platform provider.

Identification of a leak was initially dependent upon a member of the crew visually spotting the leaking slip joint packer, or worse, a spill being spotted in the ocean. Given the time taken to identify such an event, this could result in a significant environmental incident. Technology was then introduced where the packer pressure was monitored with an alarm sounding when a drop in pressure was detected. Thereby alerting the crew to activate the back-up slip joint packer (Upton 2009). However, such systems are dependent upon the manual setting of a fixed alarm point. One of the challenges faced in monitoring slip-joint packer seals, is that as the packer element wears, the pressure on the packer must be steadily increased to maintain an effective seal (Fig 1). Given this gradual rise in packer pressure, a rapid response time is dependent upon the alarm point being regularly adjusted against the operating pressure. If the manual alarm point is not constantly updated, a considerable amount of pressure and fluid from the riser could be lost before the crew are alerted, and the back-up packer engaged to stop the spill of fluid into the ocean.

Figure 1—Packer management systems installed in either a safe area or Zone 1 compliant cabinet. Both options have a small footprint of less than 2m². The Zone 1 compliant cabinet (as pictured on the right) can allow the packer management system to be placed closer to the slip joint packer assembly for a faster reaction time via shorter lengths of hydraulic line running to the cabinet. The color screen pictured on the Zone 1 cabinet shows the system configuration, pressures and status of all three packers on a floating rig or platform. In this case the primary upper packer is active (highlighted in green) and the lower back-up packer is ready for deployment (highlighted in red).

Introduction of slip joint packer management technology to prevent fluid leaks to the ocean

Given the challenge in effectively monitoring slip joint packer life on floating rigs and platforms, and the significant risk of environmental incidents resulting from failure. Focus was placed upon the development and introduction of a new advanced packer management system to prevent spills to the ocean. Rather than being simply based upon analogue pressure transducer technology, the new system is digital and designed to accelerate reaction time, immediately recognizing active slip joint packer failure and automatically activate the backup packer to prevent a spill to the ocean. The crew being alerted with both audible and visual alarms.

In addition to monitoring pressure on the primary active packer. The new packer management system also monitors the lower back up packer to ensure there is adequate pressure to successfully react in the event of primary packer failure. The system monitors the back-ups offline to ensure they are fully operational and ready for immediate deployment. This is done by trapping and constantly monitoring a small amount of pressure within the backup packer. In the event of a leak in the hydraulic control lines, the resulting drop in pressure will be picked up by the packer management system and the crew alerted with alarms indicating that a backup packer is not available in the event of primary failure.

The packer management system is installed on semi-submersible rigs, drill ships or floating platforms in either a safe area or Zone 1 cabinet (Fig 1). The Zone 1 installation cabinet is generally preferred, as it can allow the system to be positioned closer to the slip joint packer assembly for a faster reaction time. Both installation types have a small footprint of less than $2m^2$ and are fully automated with power back-up in the event of rig power failure. Color screen displays (Fig 2) provide visual information on the system status, packer pressures, together with alarm status and control panel settings. Additional repeater color display screens and audible alarms can be placed on the rig or platform in areas such as the drill floor doghouse, the subsea engineer's office or the offshore installation manager's office.

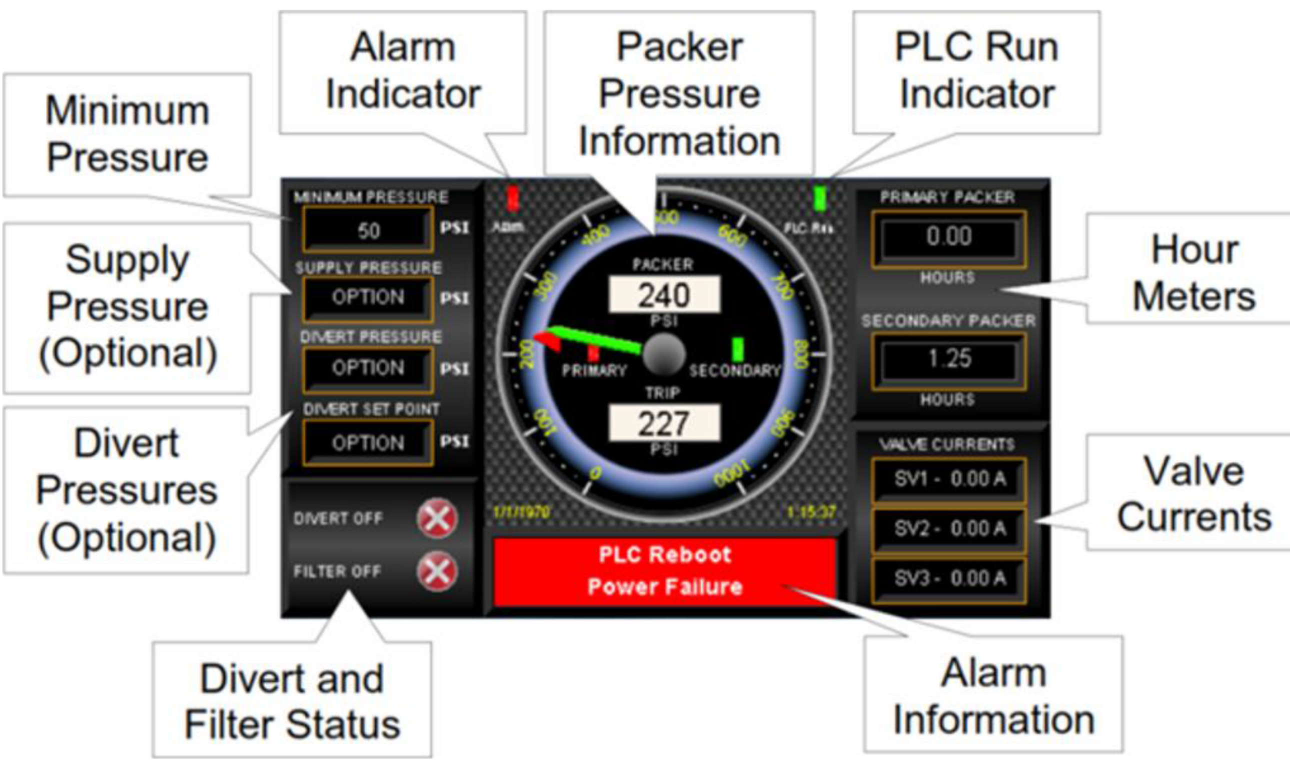

Figure 2—The packer management system display on the control panel provides visual information covering system status, packer pressure readings, power status, alarm status, settings and control parameters. Repeater displays and audible alarms can also be placed in multiple locations around the rig or platform.

One of the immediate priorities in accelerating reaction time was ensuring that an optimum alarm trip point can be set and maintained against the steadily rising slip joint packer pressure. This is crucial in ensuring a rapid response. The system monitors the active upper packer pressure and applies intelligent 'dead band' tracking to automatically adjust the alarm trip point. (Fig 3) Whenever the active packer pressure rises above the upper pressure limit window, the alarm threshold is automatically reset.

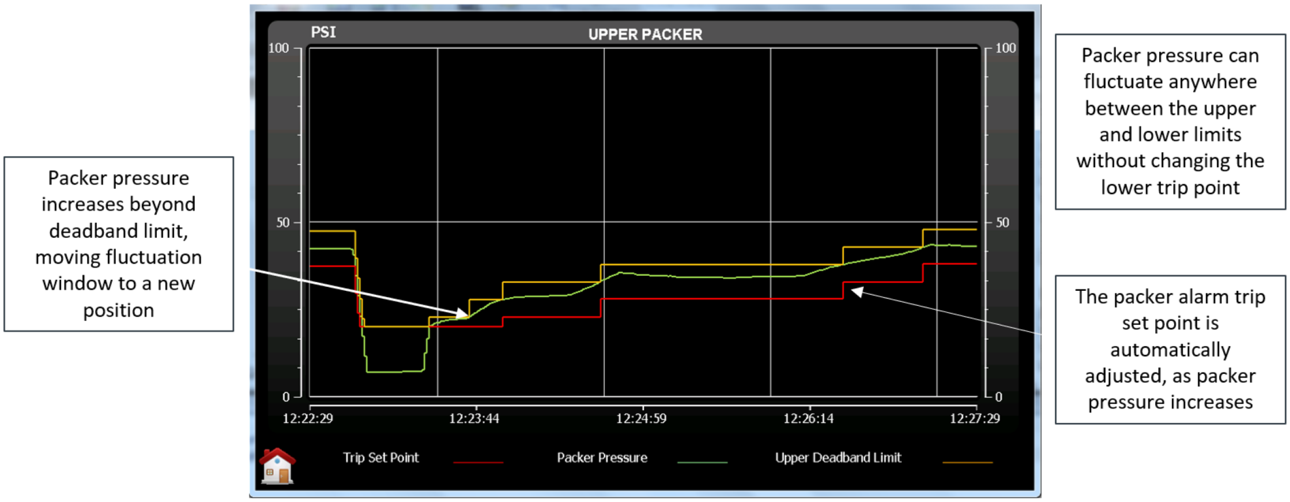

Figure 3—Chart illustrating the monitoring of pressure on the active upper slip joint packer and the intelligent system alarm settings. The packer pressure (in green) can be seen being gradually increased by the rig, to maintain an effective seal as the packer sealing element wears. The packer management intelligent 'dead band' tracking can be observed constantly re-adjusting the alarm window to ensure a rapid system response. Whenever the packer pressure rises above the upper pressure limit (in yellow) the alarm trip point (in red) is automatically re-set without the need for human intervention. With rig movement, the packer pressure can fluctuate within the upper and lower limits without changing the lower trip point and thus avoiding false alarms.

If the active packer pressure falls below the alarm trip point (Fig 4) the management system immediately actuates the backup packer to seal fluid within the telescopic riser and avert a spill. Audible and visible alarms are generated to alert the crew. The packer management system also monitors the rig divertor function. Upon seeing a divert activation the system will energize all packers to maximize environmental protection. All data is recorded by the packer management system and can be reviewed by the client in the event of an incident.

Figure 4—Chart illustrating a fall in pressure on an active slip-joint packer, as would be observed in the event of packer failure. When the pressure on the active packer falls below the alarm trip point, the back-up packer is immediately actuated with audible alarms being sounded to alert the crew and visual alarms being triggered on both the packer management panel and repeater displays in the accommodation.

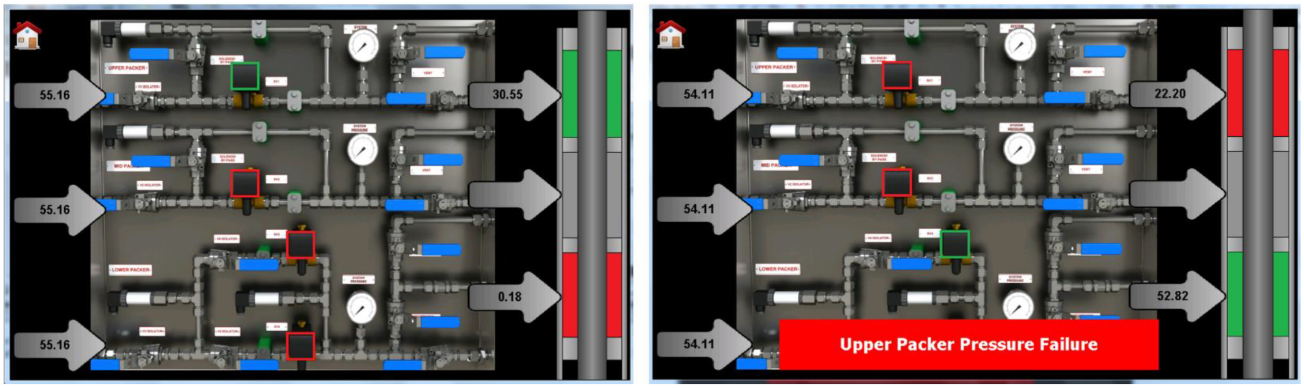

Figure 5—Illustrates the set up and monitoring of a three slip-joint packer installation on a floating rig, drill ship or platform. The right hand image depicts failure of the upper primary packer (in red) and successful automatic deployment of the lower back-up packer (in green). Audible and visual alarms are sounded to alert the crew to upper packer failure.

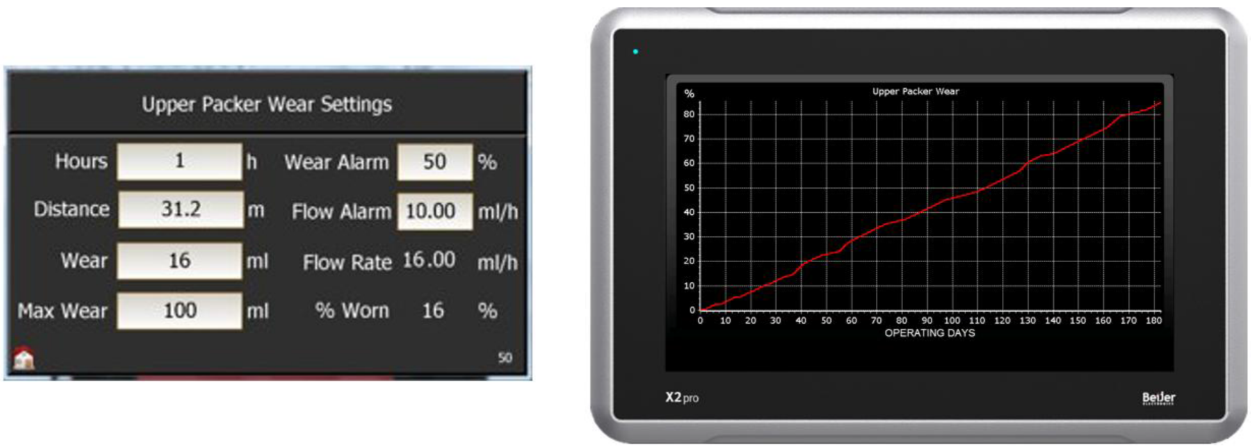

Figure 6—Illustrates the monitoring of slip joint packer wear on a floating rig, drill ship or platform. The degree of wear is determined and expressed as a percentage of packer life. Customizable alarms are set to alert the crew and clients that preventative maintenance should be carried out to avoid packer failure.

The monitoring of multiple primary and backup slip joint packers

Modern semi-submersible rigs and drill ships are increasingly equipped with multiple primary and backup slip joint packers. This allows multiple additional backup packers to be ready for deployment. If a slip joint packer leak is detected and only a single back up is available, while a fluid spill would have been averted, it would be necessary for the operator to suspend operations while the leaking packer element is replaced to ensure a further packer is available for deployment. Potentially resulting in non-productive time and project delays. Given the importance of multiple backup packers in enabling continuous operations, the latest packer management systems can simultaneously monitor multiple slip joint packers. Thus, ensuring continuous monitoring and protection. Supply pressure is monitored to all packers to ensure that they are constantly operational. If a drop in pressure on one of the backup packers is detected, audible and visual alarms are sounded to alert the crew. Allowing maintenance to be immediately carried out. Pressure drops in supply lines can indicate a leak of hydraulic fluid, and the potential for failed packer deployment.

The remote transmission of data

New technology has also been developed and applied to allow the two-way digital transmission of data from the packer management system. Given the growing trend in the deployment of rigs and platforms which either have a reduced crew compliment or are completely unmanned, there is an increasing focus on technologies which can be monitored and accessed remotely. While the packer management technology will react independently without human intervention, its status, settings and all pressure data can now be remotely accessed by the operator and service provider. Monitoring can be carried out from a remote integrated operations centre. In the event of an incident, this enables all data to be accessed and analyzed immediately by subject matter experts. Trouble shooting and software updates can be carried out remotely, thus eliminating engineer offshore trips, reducing expense and the demand for offshore bed space, which can be particular advantageous on small rigs and platforms.

Further development of the technology to monitor packer wear and proactively predict failure points

The technology which has been thus far applied has been purely reactive. While it has been shown possible to respond to packer failure near instantaneously and prevent spills, focus has also been placed on the development of new technology to more accurately predict slip-joint packer failure and enable the more effective scheduling of preventative maintenance. Thus, applying technology which is both reactive and proactive in preventing fluid spills.

While preventative maintenance schedules are widely applied to replace slip-joint packers, the rate of wear on packer elements can be unpredictable. Packer wear is seen to increase through winter storm cycles, particularly in harsh deep water weather environments such as the Northern North Sea or the tropical cyclones encountered in the offshore USA. The long-term forecasting of harsh weather events also remains a challenge from year to year. The rate of packer wear is thus not linear, and the capability to more accurately track remaining packer life is very useful in scheduling the preventive maintenance of these critical systems. This ability to better anticipate preventative maintenance could also have a positive benefit on operator/ service provider supply chain. Allowing the more efficient procurement of spare components.

Conclusion

Since its introduction, packer management technology has been installed on over 150 floating rigs, platforms and drill ships. The technology has achieved a 100% track record with no spills recorded from slip joint packers monitored by the system. While this technology has been successful in protecting against spills from all monitored packers, we are aware of environmental incidents related to leaking back up slip joint packers that have not been monitored. For this reason, we have actively pushed ahead in developing and fielding systems where all primary and back up packers are monitored, thus de-risking operations, optimizing environmental protection and reducing incidents of non productive time and related expense for operators. While the effectiveness of the packer management technology has been proven. We believe there is considerable scope to further develop this technology from being reactive to fully proactive. The currently fielded technology is focused upon immediately recognizing slip joint packer failure and deploying backup systems. There is the opportunity to further develop and field the capability to actively measuring slip joint packer wear. Thus, allowing the better scheduling of preventative maintenance, proactively avoiding slip joint packer failure and further reducing the risk of fluid discharges to the ocean.

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